Linear Algebra and Its Applications

Section 5.6: Discrete Dynamical Systems

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.6 Discrete dynamical systems

For $x_{k+1} = Ax_k$, start by expanding x_0 in an eigenbasis: $x_0 = \sum_j c_j v_j$, $Av_j = \lambda_j v_j$. Then

$$x_k = \sum_j c_j \lambda_j^k v_j.$$

This follows by induction: multiplication by A adds one power of each eigenvalue. A mode with $|\lambda| < 1$ decays; one with $|\lambda| > 1$ grows. A negative eigenvalue also reverses its mode's sign at each step. The direction of fastest decay corresponds to the smallest modulus, whereas fastest growth corresponds to the largest modulus. These discrete-time tests concern modulus, not merely the sign of an eigenvalue.

Practice problem 5.6.P1: Determine the three initial coefficients

The given eigenvectors are $v_1 = (-2, 2, 1)^T$, $v_2 = (2, 1, 2)^T$, $v_3 = (1, 2, -2)^T$. Solve $c_1v_1 + c_2v_2 + c_3v_3 = (1, 11, -2)^T$:

$$-2c_1 + 2c_2 + c_3 = 1, \quad 2c_1 + c_2 + 2c_3 = 11, \quad c_1 + 2c_2 - 2c_3 = -2.$$

Elimination gives $(c_1, c_2, c_3) = (2, 1, 3)$; substitution checks all three coordinates. Therefore $x_k = 2v_1 + (2/3)^k v_2 + 3(1/3)^k v_3$.

Practice problem 5.6.P2: Retain only the nondecaying mode

The powers $(2/3)^k$ and $(1/3)^k$ tend to zero. Thus the preceding solution tends to $2v_1 = (-4, 4, 2)^T$, rather than zero. The eigenvalue 1 preserves its component indefinitely.

Exercise 5.6.1: Use the given eigenvectors without reconstructing the matrix

Solve $c_1(1, 1)^T + c_2(-1, 1)^T = (9, 1)^T$. The coordinate equations $c_1 - c_2 = 9$, $c_1 + c_2 = 1$ give $c_1 = 5, c_2 = -4$.

(a) Apply A once to the decomposition.

$x_1 = 5(3)(1, 1)^T - 4(1/3)(-1, 1)^T = (49/3, 41/3)^T$.

(b) Apply the same rule k times.

$$x_k = 5\,3^k(1, 1)^T - 4\,(1/3)^k(-1, 1)^T.$$

At $k = 0$ this gives the specified initial vector; multiplication by A verifies the recurrence at every subsequent step.

Exercise 5.6.2: Separate the growing mode from two decaying modes

The coefficient system $[v_1\ v_2\ v_3]c = x_0$ is

$$c_1 + 2c_2 - 3c_3 = -2, \quad c_2 - 3c_3 = -5, \quad -3c_1 - 5c_2 + 7c_3 = 3.$$

It gives $c = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$. Hence

$$x_k = 2 \cdot 3^k v_1 + (4/5)^k v_2 + 2(3/5)^k v_3.$$

The last two modes decay, whereas the first grows. More precisely, $x_k/3^k \to 2v_1 = (2, 0, -6)^T$. Thus the sequence is unbounded and its direction approaches the line through v_1; its second coordinate tends to zero while its first and third grow in opposite signs.

Exercise 5.6.3: Increase predation in the owl–wood-rat model

For $p = .2$ the matrix is $A = \begin{bmatrix} .5 & .4 \\ -.2 & 1.1 \end{bmatrix}$. The characteristic polynomial is $\lambda^2 - 1.6\lambda + .63$ with roots $.9, .7$. Eigenvectors are $(1, 1)^T$ and $(2, 1)^T$, as direct multiplication verifies. Thus

$$x_k = c_1(.9)^k(1, 1)^T + c_2(.7)^k(2, 1)^T.$$

Both modes decay. Under the exercise's $c_1 > 0$ assumption, the long-term positive populations both decline with factor approaching .9 per month. The first component counts owls; the second is the number of wood rats in thousands, as in Example 1.

Exercise 5.6.4: The borderline between eventual decline and growth

With $p = .125$, the matrix has eigenvalues $1, .6$ and eigenvectors $(4, 5)^T, (4, 1)^T$. For instance $A(4, 5)^T = (4, 5)^T$ and $A(4, 1)^T = .6(4, 1)^T$. Consequently

$$x_k = c_1(4, 5)^T + c_2(.6)^k(4, 1)^T \longrightarrow c_1(4, 5)^T.$$

The populations approach constant levels rather than tending to zero or growing indefinitely. Small parameter changes can move the largest eigenvalue below or above 1, changing the prediction to decline or growth. This explains the sensitivity called an unstable equilibrium in the exercise; the equilibrium line is not a robust attracting equilibrium under model changes. Coordinate ratio $4 : 5$ means 4 owls per 5 thousand rats under the model's units.

Exercise 5.6.5: Long-term growth and the dominant population ratio

For $p = .325$, $A = \begin{bmatrix} .4 & .3 \\ -.325 & 1.2 \end{bmatrix}$. Its eigenvalues solve $\lambda^2 - 1.6\lambda + .5775 = 0$, giving $1.05, .55$. A dominant eigenvector is $(6, 13)^T$, because $A(6, 13)^T = (6.3, 13.65)^T = 1.05(6, 13)^T$. Under the stated positive dominant-coefficient assumption, x_k is asymptotic to $c_1(1.05)^k(6, 13)^T$.

Conclusion. Both populations eventually grow about 5% per step, and their coordinate ratio tends to $6 : 13$. If the prey coordinate uses the thousands convention of Example 1, this is 6 owls per 13 thousand squirrels; coordinate scaling must be retained when interpreting a physical ratio.

Exercise 5.6.6: Choose the predation rate that preserves population levels

At $p = .5$, the characteristic polynomial is $\lambda^2 - 1.6\lambda + .63$, whose roots are $.9, .7$. Both powers tend to zero, so both populations perish in this model. For constant limiting levels, require an eigenvalue 1:

$$\det(A - I) = (-.6)(.2) - (.3)(-p) = -.12 + .3p = 0.$$

Thus $p = .4$. The other eigenvalue is .6, and a 1-eigenvector is $(1, 2)^T$. Therefore the solutions tend to $c_1(1, 2)^T$.

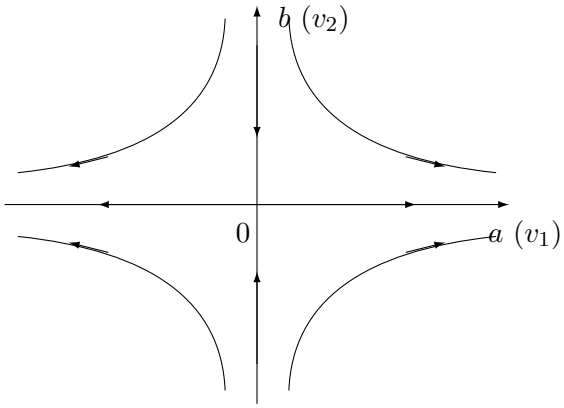
Conclusion. The preserving parameter is $p = .4$, and the limiting coordinate ratio is $1 : 2$, with prey-unit scaling interpreted as in the preceding exercise.

Exercise 5.6.7: A saddle with one attracting and one repelling direction

(a) Since $3 > 1$ but $1/3 < 1$, the origin is a saddle point.

(b) The line spanned by $v_1 = (1, 1)^T$ is repelling, with factor 3 each step. The line spanned by $v_2 = (-1, 1)^T$ is attracting, with factor $1/3$. These are also the greatest repulsion and attraction directions, respectively.

(c) In eigen-coordinates $x = av_1 + bv_2$, the update is $(a, b) \mapsto (3a, b/3)$. The sketch uses those coordinates: horizontal arrows point away from zero and vertical arrows toward zero. Off-axis trajectories approach the repelling axis while moving outward. The curves are schematic connections of discrete points, not continuous-time trajectories; ab remains constant here.



Exercise 5.6.8: Two stable modes and one unstable mode

The eigenvalues from Exercise 5.6.2 are $3, 4/5, 3/5$. A mode expands along $(1, 0, -3)^T$, while the two-dimensional span of the other eigenvectors is attracted to zero. The fastest contraction has the smaller modulus $3/5$, along $(-3, -3, 7)^T$; the greatest repulsion has modulus 3, along $(1, 0, -3)^T$.

Conclusion. The origin is a saddle. The initial vector in Exercise 2 has coefficient $c_1 = 2$, so its trajectory is repelled. Only initial vectors in the stable plane ($c_1 = 0$) tend to zero.

Exercise 5.6.9: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{17}{10} & -\frac{3}{10} \\ -\frac{6}{5} & \frac{4}{5} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(\lambda - 2)(2\lambda - 1)}{2} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{1}{2},\ v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}; \qquad \lambda = 2,\ v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **saddle point**. The attracting direction is $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ (factor $\frac{1}{2}$), and the repelling direction is $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ (factor 2).

Exercise 5.6.10: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{3}{10} & \frac{2}{5} \\ -\frac{3}{10} & \frac{11}{10} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(2\lambda - 1)(10\lambda - 9)}{20} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{1}{2},\ v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{9}{10},\ v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **attractor**. The greatest attraction is along $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, for the smaller eigenvalue $\frac{1}{2}$. There is no repelling eigendirection.

Exercise 5.6.11: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{2}{5} & \frac{1}{2} \\ -\frac{2}{5} & \frac{13}{10} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(5\lambda - 4)(10\lambda - 9)}{50} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{4}{5},\ v = \begin{bmatrix} 5 \\ 4 \end{bmatrix}; \qquad \lambda = \frac{9}{10},\ v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **attractor**. The greatest attraction is along $\begin{bmatrix} 5 \\ 4 \end{bmatrix}$, for the smaller eigenvalue $\frac{4}{5}$. There is no repelling eigendirection.

Exercise 5.6.12: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{1}{2} & \frac{3}{5} \\ -\frac{3}{10} & \frac{7}{5} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(5\lambda - 4)(10\lambda - 11)}{50} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{4}{5},\ v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{11}{10},\ v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **saddle point**. The attracting direction is $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ (factor $\frac{4}{5}$), and the repelling direction is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (factor $\frac{11}{10}$).

Exercise 5.6.13: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{4}{5} & \frac{3}{10} \\ -\frac{2}{5} & \frac{3}{2} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(5\lambda - 6)(10\lambda - 11)}{50} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{11}{10},\ v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{6}{5},\ v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **repeller**. The greatest repulsion is along $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$, for the larger eigenvalue $\frac{6}{5}$. There is no attracting eigendirection.

Exercise 5.6.14: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{17}{10} & \frac{3}{5} \\ -\frac{2}{5} & \frac{7}{10} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(10\lambda - 13)(10\lambda - 11)}{100} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{11}{10},\ v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{13}{10},\ v = \begin{bmatrix} -3 \\ 2 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **repeller**. The greatest repulsion is along $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$, for the larger eigenvalue $\frac{13}{10}$. There is no attracting eigendirection.

Exercise 5.6.15: A stochastic model with a nonzero limiting vector

The columns of A sum to 1. The given $v_1 = (.1, .6, .3)^T$ has $Av_1 = v_1$. Eigenvectors for $.5, .2$ are $v_2 = (2, -3, 1)^T$ and $v_3 = (-1, 0, 1)^T$. Solving $c_1v_1 + c_2v_2 + c_3v_3 = (0, .3, .7)^T$ gives $c_1 = 1, c_2 = 1/10, c_3 = 3/10$. Thus

$$x_k = v_1 + \tfrac{1}{10}(.5)^k v_2 + \tfrac{3}{10}(.2)^k v_3.$$

At zero the coordinates are $(.1 + .2 - .3, .6 - .3, .3 + .1 + .3)$, as required. The last two modes tend to zero, so $x_k \to v_1$. The limiting proportions are $10\%, 60\%, 30\%$.

Exercise 5.6.16: The full Hertz three-location solution

In City Airport, Downtown, Metro Airport order, the matrix from Section 4.9 is

$$A = \frac{1}{100}\begin{bmatrix} 90 & 1 & 9 \\ 1 & 90 & 1 \\ 9 & 9 & 90 \end{bmatrix}.$$

Exact calculation gives three eigenpairs:

$$1,\ (91, 19, 99)^T; \qquad .89,\ (-1, 1, 0)^T; \qquad .81,\ (-1, 0, 1)^T.$$

The eigenvalues are distinct, so their eigenvectors form a basis. Therefore the general solution is

$$x_k = c_1(91, 19, 99)^T + c_2(.89)^k(-1, 1, 0)^T + c_3(.81)^k(-1, 0, 1)^T.$$

The constants are determined by the initial fleet distribution. For total fleet size N, adding the coordinates gives $N = 209c_1$, so the limit is $(N/209)(91, 19, 99)^T$. At $N = 2000$ the downtown limit is $2000(19)/209 \approx 181.818$ cars. This recovers the steady-state interpretation while providing the requested general dynamics.

Exercise 5.6.17: A two-stage growing population with all four requested graphs

(a) Translate births and survival into matrix entries.

Juveniles next year are born to this year's adults, while next year's adults are surviving juveniles plus surviving adults. Hence

$$A = \begin{bmatrix} 0 & 1.6 \\ .3 & .8 \end{bmatrix}.$$

(b) Find growth and the eventual ratio.

The characteristic polynomial is $\lambda^2 - .8\lambda - .48$, with roots $1.2, -.4$. A dominant eigenvector is $(4, 3)^T$; a second is $(-4, 1)^T$. The growing mode therefore has factor 1.2, a 20% annual growth rate, and limiting juvenile/adult ratio 4/3. For a nonzero nonnegative initial population the growing coefficient is positive: solving $x_0 = c_1(4, 3)^T + c_2(-4, 1)^T$ gives $c_1 = (j_0 + 4a_0)/16 > 0$.

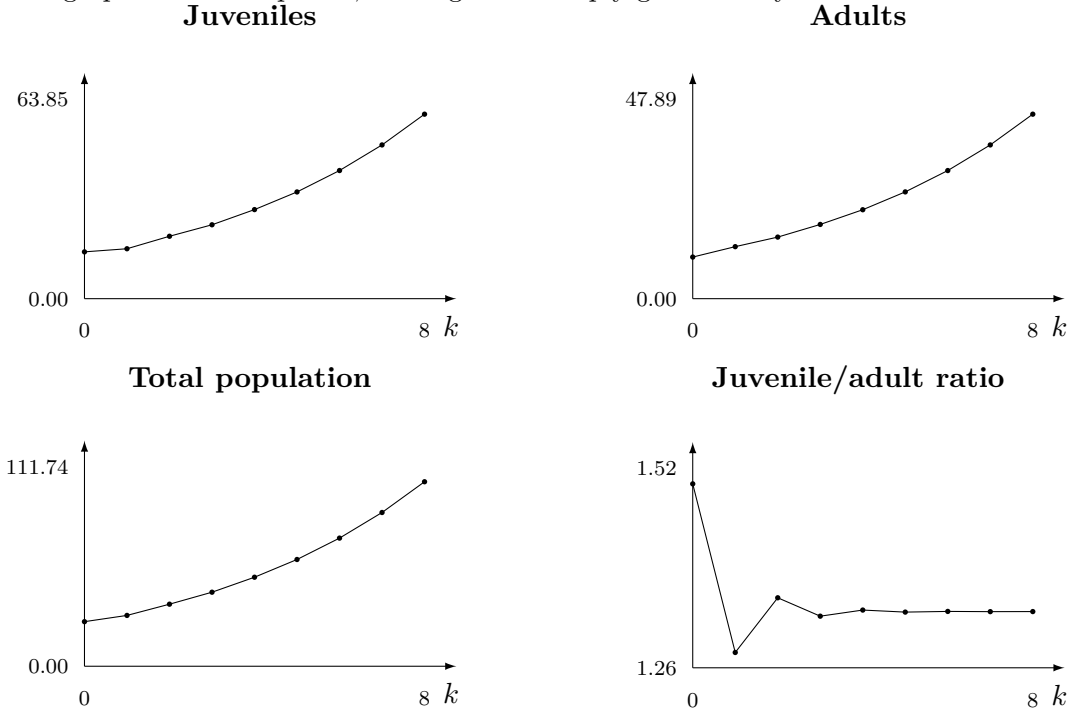
(c) Calculate the eight-year trajectory and plot it.

Beginning at $(15, 10)^T$, the actual iterates are

k	j_k	a_k	j_k/a_k
0	15.000000	10.000000	1.500000
1	16.000000	12.500000	1.280000
2	20.000000	14.800000	1.351351
3	23.680000	17.840000	1.327354
4	28.544000	21.376000	1.335329
5	34.201600	25.664000	1.332668
6	41.062400	30.791680	1.333555
7	49.266688	36.952064	1.333259
8	59.123302	44.341658	1.333358

.

The four graphs use these points; line segments simply guide the eye between successive integer years.



The ratio oscillates toward 4/3 because the secondary eigenvalue is negative. It is already close by years 3–4 and differs by less than .001 from 4/3 by year 5. The underlying relative mode shrinks by $|-.4/1.2| = 1/3$ each year. A complete plotting program, including the requested total and ratio panels, is:

```
import numpy as np
import matplotlib.pyplot as plt
A = np.array([[0.,1.6],[.3,.8]])
x = np.array([15.,10.])
values = [x.copy()]
for k in range(8):
    x = A @ x
    values.append(x.copy())
v = np.array(values)
series = [v[:,0],v[:,1],v.sum(axis=1),v[:,0]/v[:,1]]
```

```
labels = ['Juveniles','Adults','Total','Juveniles/adults']
fig, axes = plt.subplots(2,2)
for ax, y, label in zip(axes.flat, series, labels):
    ax.plot(range(9), y, 'o-')
    ax.set(xlabel='Year', ylabel=label)
fig.tight_layout()
plt.show()
```

Exercise 5.6.18: A three-stage buffalo model

(a) Construct the stage matrix.

Only adults give birth. Calves survive into the yearling stage and yearlings into adulthood. Thus

$$A = \begin{bmatrix} 0 & 0 & .42 \\ .60 & 0 & 0 \\ 0 & .75 & .95 \end{bmatrix}.$$

(b) Compute the dominant mode and its composition.

The characteristic polynomial is $\lambda^3 - .95\lambda^2 - .189$. Its real positive root is $\lambda_1 \approx 1.104834392$; the other two roots are complex conjugates of modulus less than 1. Since $\lambda_1 > 1$, the nonzero nonnegative population grows asymptotically by factor 1.104834392, about 10.483439% per year. Normalize the dominant eigenvector to 100 adults. The first row of $Av = \lambda_1 v$ gives $c = 42/\lambda_1$, and the second gives $y = .6c/\lambda_1$. Therefore the eventual composition per 100 adults is approximately

$$(c, y, a) = (38.014747, 20.644586, 100).$$

Substituting these formulas into the third row reproduces the characteristic equation; this checks the eigenvector normalization.

Show that every nonzero nonnegative initial population has a growing mode.

An initial state could in principle have zero coefficient in a particular eigenvector direction, so we must verify that the dominant coefficient is positive here. A useful row multiplier is the positive left eigenvector

$$w = \begin{bmatrix} .45/\lambda_1^2 \\ .75/\lambda_1 \\ 1 \end{bmatrix}.$$

Indeed direct multiplication gives

$$w^T A = \begin{bmatrix} .45/\lambda_1 & .75 & .189/\lambda_1^2 + .95 \end{bmatrix} = \lambda_1 w^T.$$

The last equality uses the characteristic equation, divided by λ_1^2: $\lambda_1 = .95 + .189/\lambda_1^2$. All three entries of w are strictly positive because $\lambda_1 > 0$.

The three distinct eigenvalues give an eigenvector expansion $x_0 = c_1 v_1 + c_2 v_2 + c_3 v_3$, allowing complex coefficients for the conjugate pair. For $j = 2, 3$, calculate the same scalar in two ways:

$$w^T A v_j = \lambda_1 w^T v_j = \lambda_j w^T v_j.$$

Since $\lambda_j \neq \lambda_1$, this forces $w^T v_j = 0$. Multiplying the expansion of x_0 by w^T therefore isolates

$$c_1 = \frac{w^T x_0}{w^T v_1} > 0.$$

The numerator is positive whenever x_0 has nonnegative entries and at least one positive entry; the denominator is positive because v_1 has the positive components computed above. Consequently

$$\frac{x_k}{\lambda_1^k} = c_1 v_1 + c_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k v_2 + c_3 \left(\frac{\lambda_3}{\lambda_1}\right)^k v_3 \longrightarrow c_1 v_1.$$

Both remaining ratios have modulus less than 1, so their powers tend to zero. Thus the dominant mode really is present for every nonzero nonnegative initial population, justifying both the growth rate and the eventual stage proportions claimed above.